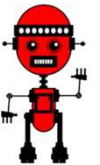
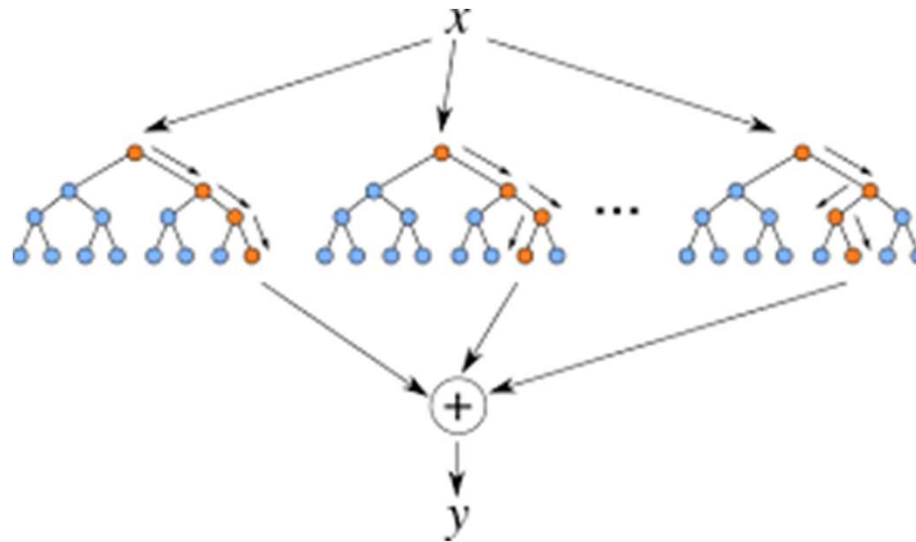


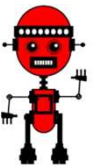
Random Forest Machine learning Algorithm



CONTENT



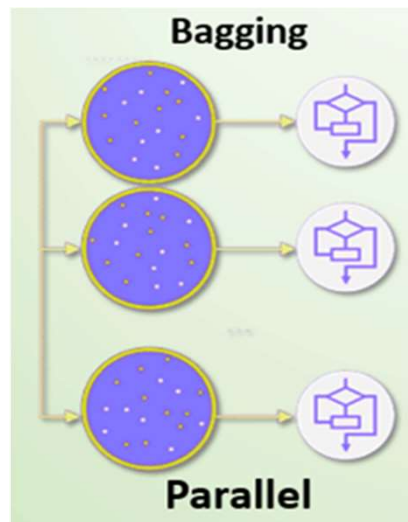
- Ensembling Learning
- Concept of Random Forest
- Random Forest Algorithm
- Hyperparameters



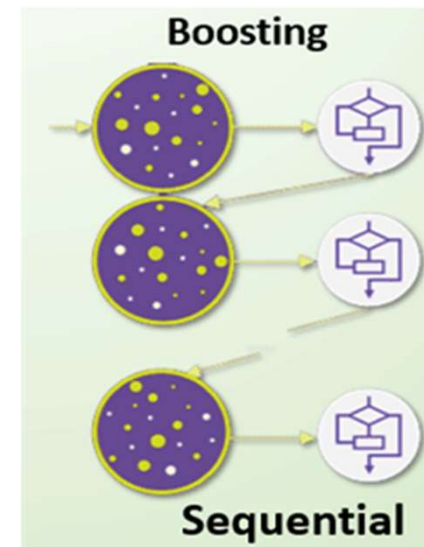
Ensemble Methods

Process of combining multiple classifiers to solve a complex problem and to improve the performance of the model.

Bagging : Creating a different training subset from sample training data with replacement. The final output is based on majority voting.

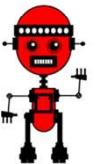
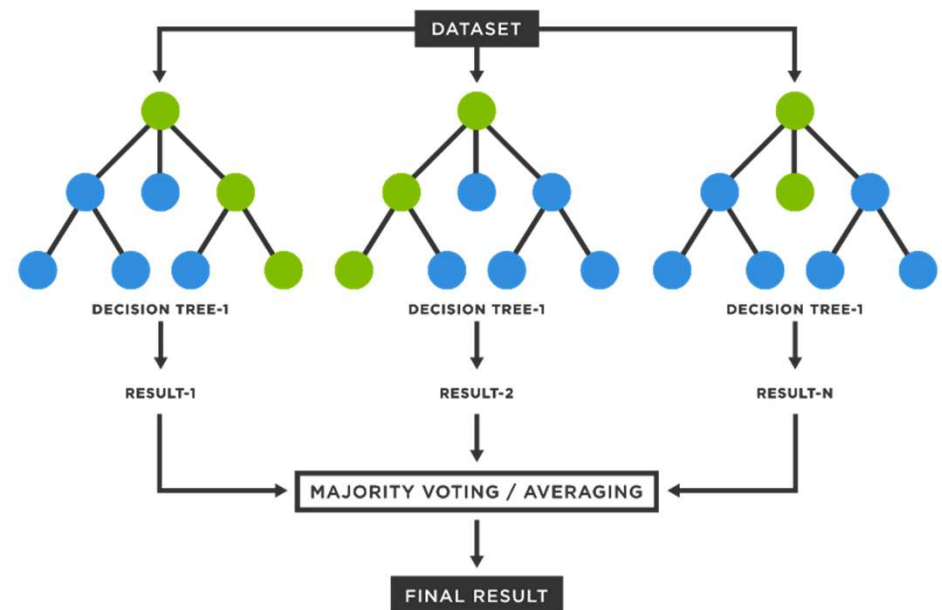


Boosting : Combining weak learners into strong learners by creating sequential models such that the final model has the highest accuracy. Example: ADA BOOST, XG BOOST.



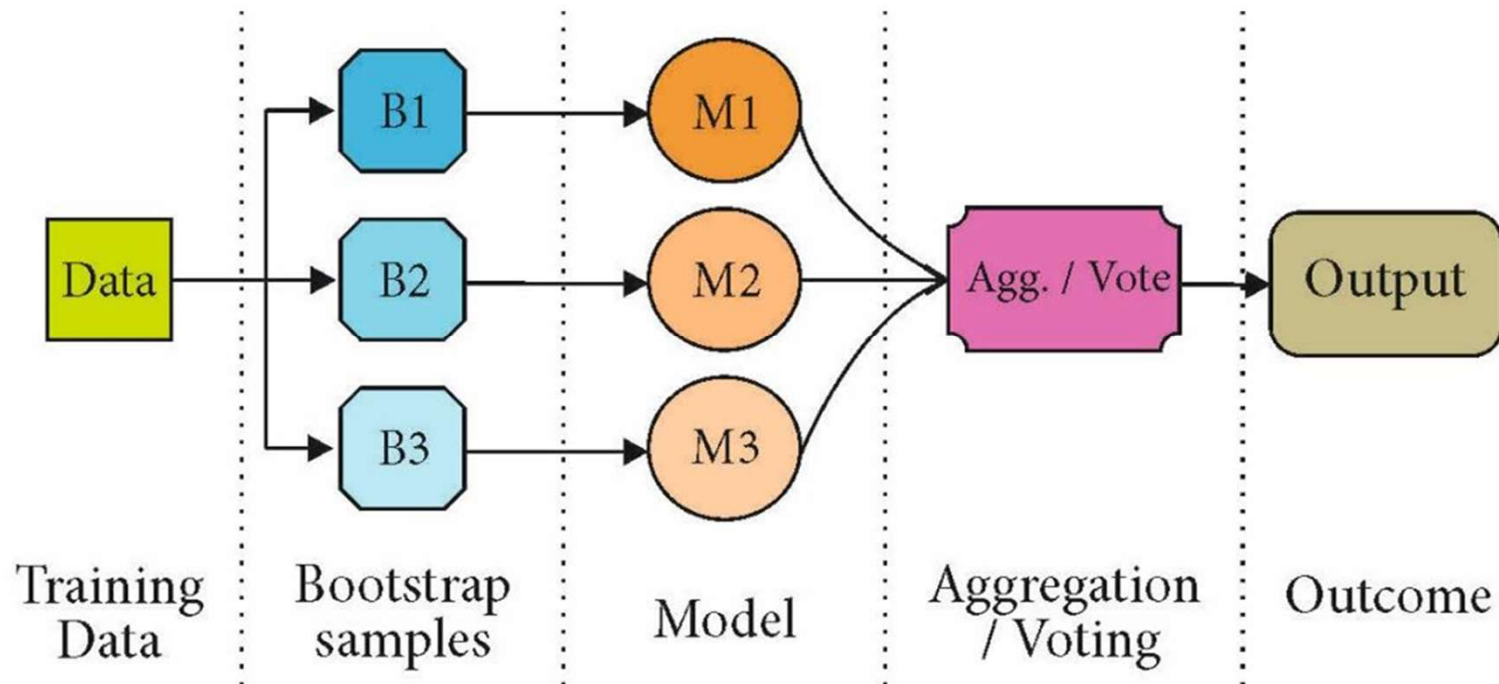
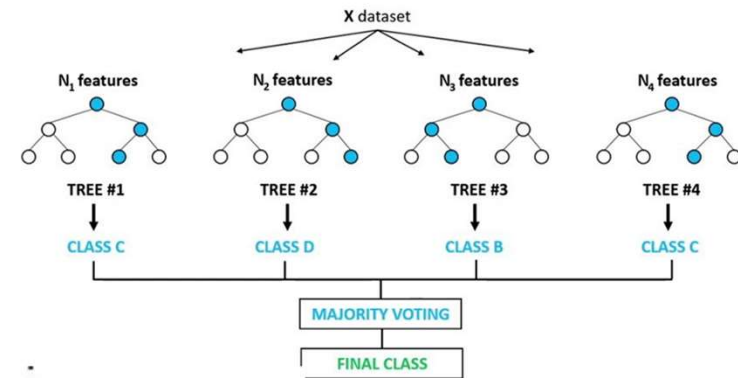
Concept of Random Forest

- Supervised Machine learning algorithm
- Made up of multiple decision trees
- Used for both regression and classification problems
- Based on bagging Ensemble learning



Random forest algorithm

- Utilizes both bagging and feature randomness
- Create an uncorrelated forest of decision trees
- Based on voting, final output is predicted.



Hyperparameter of random forest

Node size

The number of trees

Number of features sampled

1. `n_estimators`– number of trees the algorithm builds before averaging the predictions.
2. `max_features`– maximum number of features random forest considers splitting a node.
3. `mini_sample_leaf`– determines the minimum number of leaves required to split an internal node.

Preferred --

Random Forest is suitable for situations when we have a large dataset, and interpretability is not a major concern.

Benefits

- Normalization is not required.
- Reduce the risk of overfitting, bias, and overall variance.
- Predicts output with very high accuracy